

The Second Storm Doesn't Find a Pristine Forest

Why Near-Term AI Augments Labor Rather Than Replacing It

Yakov Pyotr Shkolnikov*
Independent Researcher

July 16, 2026 (analysis cutoff: July 2026)

Abstract

The prevailing forecasts say AI will displace much of office and physical work within a few years, and firms are already budgeting their headcount against them. Those forecasts mistake which work AI automates. What today's models do reliably is the work with a clear right or wrong answer, the kind whose errors are cheap to catch. Most of that was automated long ago, by the office software and factory machines of the last century, and aggregate employment did not fall. What is left is the work that turns on judgment: deciding what the task is and whether the result is right. Generative AI has bought fluency, not judgment: a larger model writes more convincingly but is no better at telling a right answer from a merely plausible one, most likely because it learns from finished work and not the reasoning. Physical work asks for that same judgment and, on top of it, the real-time adaptation to a constantly changing world that no fixed program supplies, which is why a general-purpose robot is a far riskier bet than the specialized machines already on the line. To automate the rest, a firm has to wrap the model in a harness, expensive to build and to run, that in the near term adds more human labor instead of cutting it. On the three-to-five-year horizon, little evidence supports large aggregate displacement: the technology makes the work better, and the people who do it more capable, not fewer.

Keywords: artificial intelligence; automation; labor displacement; augmentation; productivity; robotics; task-based framework.

JEL: J23, J24, O33, E24, D24.

Automation advocates have a long tradition of claims that a new technology will rapidly displace human labor, and AI proponents are no exception. This displacement prediction is tempting, because, while a contingent still calls AI a failed technology, it clearly works and helps most along the ragged edge of what it can do, especially for personal productivity. Yet the immediate labor

*The author works on agentic AI and is not compensated for this paper; its case cuts against the near-term displacement narrative that would most benefit such work, and rests on the cited literature.

reductions have not materialized. This paper does not attempt an analysis of AI's macroeconomic impact, nor does it claim that no displacement will occur. It takes a narrower view: where the current generation of AI capability stands, set against a modern workplace that a century of ruthless optimization has already pared down, leaving little immediate market for further labor displacement (Figure 1). A claim about the near-term aggregate is not a claim that the adjustment is painless or evenly shared: earlier automation and trade held total employment roughly steady while reshaping its composition, thinning the exposed occupations and straining the wages, communities, and unions built around them [1, 2]. How those gains and losses are distributed is a separate question this paper does not take up, though the mechanism here, that standardized work goes first, bears on it.

The displacement forecasts ignore the nature of the tasks that make up modern work. A task is cheap to automate when its result has a *bounded, cheaply-checkable error surface*: the code compiles or it does not, the ledger reconciles or it does not. Where the surface is bounded and the check is cheap, fluency alone runs the task, and almost all such work was standardized and automated even before generative AI. Where it is not, the work still relies on human *judgment*: deciding what to do and whether the result is right. In knowledge work that is the bulk of what remains. Physical work adds *adaptation* on top of judgment, carrying the action out in real time in a world that keeps presenting what no policy was trained on, where the error plays out before any check could run. Judgment runs through both.

Why the machine lacks judgment we can only hypothesize from how the models are trained: pretraining runs on finished output, the settled products of judgment and never the reasoning that produced them [3], so scale improves fluency, not judgment. Recent reasoning models that train on process, not only output, do not refute this: they gain only where a cheap external check can score the steps, as in mathematics and code with tests. The intent it would need is developed only in the act, as in writing, and discarded before the output is recorded, so more data cannot recover it [4, 5].¹ Adaptation fails for a settled reason instead: the body and the open world defeat a policy tuned on recorded demonstrations [11, 12].

The job that people do is more than checking the output or writing a vague prompt. The messy work still has to be mapped into well-specified tasks, and the objective itself still has to be set. A natural-language instruction specifies neither precisely, the reason formal notation exists at all: prose too readily admits statements whose imprecision is not obvious [13]. So the machine can produce a fluent artifact that is not the one that was wanted. The economics of predictive AI and automation already left the objective with the human [14]. Generative AI does not address it either.

The tools take over some tasks and speed up others. The standing counterargument is that a harness of scaffolding, retrieval, tools, and checks built around the model can extend what it automates. It can, but slowly, and at two costs, both large. Building the harness takes skilled people working alongside the very work it is meant to automate, then supervising what it produces. Running it

¹The alignment stages that might add intent are tiny next to pretraining: tens of thousands of demonstrations and preference comparisons, by some methods a thousand [6], against pretraining corpora of trillions of tokens [7, 8]. They too consist of finished responses and preferences over outputs, not the process of authoring an objective; and reinforcement learning from human feedback, the stage most often credited with adding judgment, fits a reward model to aggregate preference, which shapes agreeableness and, under optimization pressure, trades against correctness [9, 10].

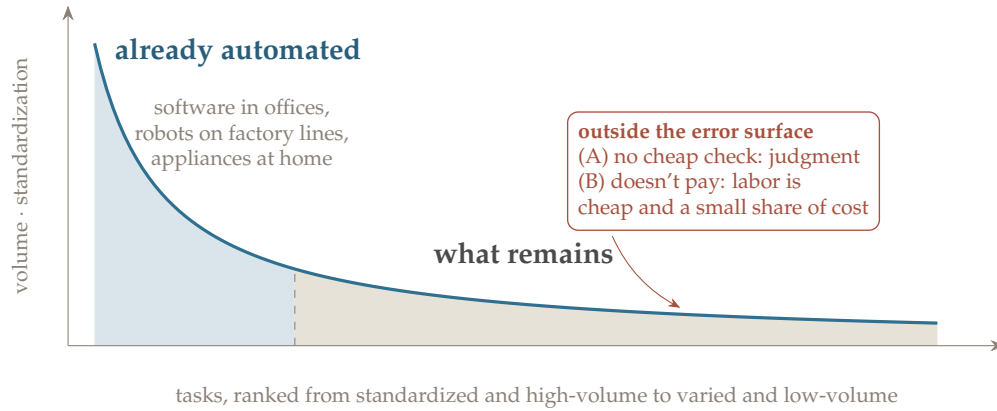


Figure 1: Schematic. Automation took the head, the work inside a bounded, cheaply-checkable error surface; the long tail stays with people, on two grounds.

means looping to manufacture a check, which consumes tokens far out of proportion to the result.

1. Office work

1.1 The routine is already automated

The computer age, as Robert Solow quipped in 1987, was visible everywhere but in the productivity statistics [15]. Those gains arrived only about a decade later, once firms had reorganized their work around the machines [16, 17]. Computers took the routine, codifiable tasks and left the non-routine ones to people [18]. The very jobs Frey and Osborne’s 2013 study rated near-certain to be automated within a decade or two [19], tax preparers at 99% and paralegals at 94%, grew in number even as their software matured; where teller and bookkeeping work did fall, it tracked older automation, the teller to ATMs and online banking and the bookkeeper to accounting software, not AI [20]. More than a decade on, that displacement has not come.

The work these forecasts imagine AI replacing largely no longer exists; earlier automation took it long ago. Yet the studies behind the forecast rate AI against standard job-task lists, chiefly the government’s O*NET inventory, as Frey and Osborne did in 2013. Those lists still name obsolete office work, from adding machines to microfilm (Table 1); the inventory records every task a role has ever included, and a clerk today does few of them by hand. Airline reservation systems took fare-and-booking work, Freddie Mac’s Loan Prospector took first-pass mortgage underwriting, e-discovery software took document review, and electronic execution took floor trading [21–24]. Through cycles of standardization, big data, robotic process automation, and industry-specific software, automation took over much of the repetitive office work, and most of it before generative AI arrived.

Table 1: Four occupations Frey and Osborne rated near-certain for computerization in 2013 (probability P), their US employment in May 2013 and May 2025, and a task the O*NET inventory still lists. Today’s AI-exposure studies, a recent MIT evaluation among them [25], score against that same inventory. Employment [20]; task text [26].

Occupation (SOC)	P , 2013	Empl. 2013	Empl. 2025	A task the inventory still lists
Bookkeeping and accounting clerks (43-3031)	0.98	1,586,380	1,373,680	“Operate 10-key calculators, typewriters, and copy machines...”
Paralegals and legal assistants (23-2011)	0.94	271,320	392,880	“...maintain documents in paper or electronic filing systems”
Tax preparers (13-2082)	0.99	67,810	76,480	“Compute taxes... using adding machines or personal computers”
Tellers (43-3071)	0.98	527,680	329,480	“...typing, filing, and microfilm photography”

1.2 Fluency, not judgment

What remains is the work colloquially described as “just figure it out.” Take even writing this research paper. Every step, from selecting a topic, to reviewing the research, to forming a thesis, to putting it into language, is a judgment, not only of how to do it but of what to do and of what a good result looks like. Assigned to AI at the outset, this paper failed. It helped with search, drafting, figure generation, and even the bulk of the wordsmithing, but it did not do the work of writing the paper; it was a useful tool for writing one.²

The tasks that existed before automation (Figure 2) almost invariably lay in the low-judgment, high-fluency corner: once the process was figured out and the volume was high enough to write a standard operating procedure, fluency alone controlled the speed of the output, and that is where the erosion of white-collar labor hit the most. The remaining work is closer to the judgment axis across almost every industry. In software engineering, the part of the work closest to fluency over the years has been automated by higher-level languages, large libraries, and configurable products; move away from that corner and there is a small region that stayed high-volume but low-automation, and over the last three years that is where AI has helped developers most. Above it lies the majority of the remaining white-collar work. The capability frontier others have mapped as jagged against apparent task difficulty [27] is straighter against this one, though never perfectly: much of what looks like unpredictable success is whether the result has a cheap check.

The tools are still valuable. They take over particular tasks and help with many more, and the value of helping workers this way has long exceeded the value of replacing them [18, 28]; most real use is augmentation, not automation [29]. The controlled studies show real gains. Support agents resolved 14% more issues an hour, the newest workers gaining about 34% and the most

²The thesis, the understanding behind it, the initial draft, and responsibility for the claims are the author’s. It was substantially researched with AI help and first drafted, by the author, under the present title in April 2026, then revised over the months since, driven by his own review and further research; the paper now incorporates six further essays he wrote on the topic and the companion calculator referenced later. Producing it ran on self-developed research and writing harnesses, for search, verification, drafting, and editing, and the tools automated much of that work throughout. What they could not supply was judgment: only about a quarter of the wording survives from that first draft (word-set Jaccard 0.25 over some 190 revisions), reflecting the understanding the tools lacked and the significant errors in their drafts the author had to catch.

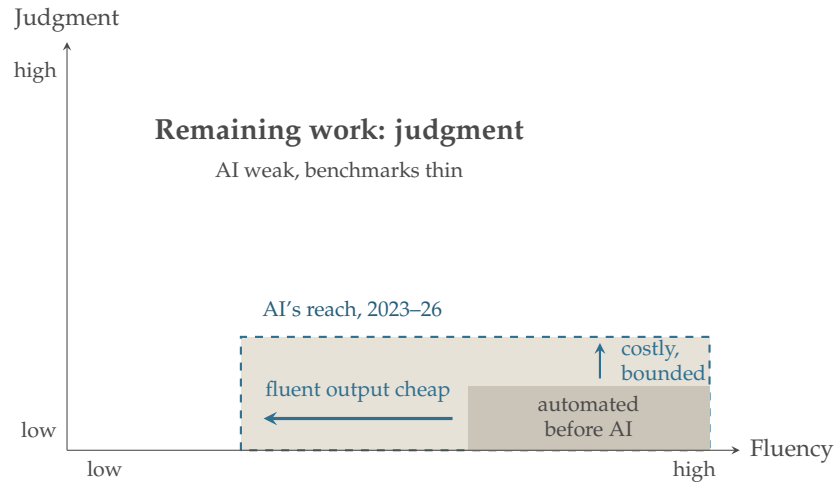


Figure 2: Schematic. Automation cleared a box of low-judgment, high-fluency work (solid). Generative AI expands it unevenly (dashed): mostly leftward, as producing fluent output becomes cheap, and only slightly upward, as the judgment it can handle rises a little, bought with a costly harness. The majority of the work sits above, where judgment dominates and AI is weakest. The horizontal axis is how much a task is the production of well-formed output.

experienced almost none [30]. Professionals wrote 40% faster on short writing tasks, and scored higher [31]. Developers finished a defined coding task 55.8% faster, where a test shows at once whether the code runs [32].

1.3 The cost of automation

But the fluency may paradoxically backfire. A randomized 2025 trial of experienced developers working in codebases they knew deeply, a setting that favors the human, found them about a fifth slower with AI, though they believed themselves faster [33], and a large 2026 study found debt across thousands of projects, almost a quarter of it unfixed long after the tools moved on [34]. The reversal is not confined to software: in a field experiment with management consultants, those who used AI on a task beyond its frontier were 19% less likely to reach the right answer than those who used none [27]. The verification cannot be skipped, and it is a task people are poorly built for: a reviewer working through a high volume of mostly-correct output meets the vigilance problem, the long-noted irony that automating the routine leaves the human a monitoring job that erodes their attention [35, 36]. The model’s mistakes compound it, because they are not random: it handles each unclear case the usual way, so errors concentrate in the unusual ones, exactly the cases that most need a person and that a fatigued monitor is least likely to catch. And the benchmarks that would tell us otherwise cover this region least, and their design routinely undermines the capability claims drawn from them [37].

Structured data should be automation’s easy case: the numbers are exact, and a query either runs or it does not. Yet even here, automating the analysis was not a one-off cost. A model set to write its own metric definitions inside Anthropic produced plausible ones that encoded the ambiguities

they were meant to remove, and a smaller human-curated set beat them on the company's own evaluations [38]: a person had to own the definitions from the start. And ownership did not end at setup: as the business shifted, the untended layer fell from 95% to 65% accuracy in a month. If a person cannot be removed even here, on governed data with exact answers, the case for removing them from looser work is weaker still.

One could argue that with a good enough harness and enough prompt engineering this work can be automated too. It is unlikely to happen fast enough to drive large displacement in the next few years, and the reason is the two costs of the harness. The first is building it. Harnesses for software engineering were expensive to develop, but at least the code's correctness could be checked and the people who wrote them were the subject-matter experts themselves. For most other work, neither holds, and even the coordination between the developers and the people who hold the requirements will delay it. Worse, in the near term it *raises* the demand for skilled labor rather than lowering it: the same worker carries the day job, the build, and the review at once.

The second cost is running it. Simulating judgment through a verification loop is the unnatural core of agentic use, and it generates far more tokens, and far more cost, than a person doing the same work: in agentic coding, the one domain measured, an agentic run already consumes on the order of a thousand times the tokens of ordinary generation, an illustrative magnitude rather than a settled constant [39]. The usual hope, falling token prices, does not provide an escape, because as the price per token drops these workflows consume disproportionately more tokens per task, so cost per task has risen, not fallen; cheaper tokens buy more looping, not a cheaper result. Nor does dropping to a cheaper model help: on a quality-adjusted basis the low-cost models deliver roughly a third the intelligence per token of the frontier ones, so running a weaker model more times cannot recover the gap [40].

1.4 The impact so far

Step back from the task to the firm, and the aggregate has barely moved. In a survey of nearly 6,000 executives, about nine in ten reported no change to employment or productivity over three years [41], and a 2025 study across thousands of workplaces finds the effect on pay and hours still too small to detect [42]; an industry survey of enterprise pilots reports the same [43]. Measured task by task rather than by whole occupation, the exposed share is far smaller: Frey and Osborne's occupation-level method puts 47% at risk [19], while the more credible task-level method puts it near 9% [44].

Even once AI is integrated, the payoff is slow to appear. A new team captures it almost at once, while an established firm first carries its legacy systems and reviews everything the tools produce, so its return is delayed by years. The lag is organizational as much as technical: even with a good harness, the gain appears only as the firm's roles, reviews, and hand-offs reorganize around the tools, and that reorganization is slow, the same delay that held the computer's own gains out of the productivity statistics for a decade. A companion interactive calculator works this through for software engineering.³

³https://yakovshkolnikov.com/employment_july_2026.html

One number does move. Young workers in highly AI-exposed jobs, the entry-level and standardized ones, have lost about 16% of their employment relative to older workers in the same firms, a gap that survives controls for the wider hiring slump [45]. Even Anthropic’s own labor study, weighting automation over augmentation, reports the same shape, no systematic rise in unemployment among the most-exposed workers through 2026, and a barely-significant 14% dip in the entry-level job-finding rate in high-exposure roles [46]. A St. Louis Fed analysis ties roughly a third of the rise in youth unemployment to AI exposure in what employers ask for, not to firms automating jobs, with weak hiring the larger share [47]; other work reports the same pattern [48]. Entry-level roles hold the most standardized tasks, the first to be automated.

But the number may register a reaction to the forecast as much as a result of it: firms cutting junior hiring on the belief that the tools will soon do the work, not because they already do. If so it can reverse. AI does raise productivity, and cheaper output can pull juniors back, while seniors stay for the institutional knowledge the tools cannot hold; cutting the juniors meanwhile thins the pool the next seniors come from. The open question is whether automation stops at the standardized tasks or goes on to take the judgment work too. This paper’s three-to-five-year call assumes it stops.

2. Service and physical work

2.1 The routine is already automated

The physical world has been through the same century of standardization. Homes were automated by specialized machines: a washer, dryer, dishwasher, and coffee maker already work as robots, only without the AI, and alongside other changes they cut the average week of housework from about fifty-eight hours in 1900 to eighteen by 1975 [49]. Industry was automated by fixed robots that took the structured, high-volume work decades ago, on cycle time and payback math rather than anything like AI, and more than four million are at work today [50]. Installations are still rising, but the growth has moved off the saturated automotive floor, now flat, into newer sectors, with food-industry robots up about 30% in the US in 2025 [51]. Service standardized the same way, from the self-service fuel pump to the boarding-pass kiosk. Even the trades put power tools and prefabricated parts in place of hand labor. Here too the routine work went before generative AI.

2.2 Judgment, and adaptation on top

The forecast of displacement of humans by humanoid robots rests on a far harder bet than industrial robotics, which works only where the setting is fenced and fixed: a welding cell, or a semiconductor fab whose automated material handling runs a whole floor unattended. The open world is neither, and the bet is a general-purpose machine meant to walk into any human setting and take the long tail of varied work that resists standardization. Even the robot vacuum, the household success, still needs a person to free it when it jams and empty it when it fills.

Physical work needs the same judgment as office work. As in the office, the machine can take what is completely standardized and purely repetitive. But it leaves the person not only to decide what the job is and whether it is done, but also to carry that decision out in real time, in a space that is

constantly changing.

Humans, and even simple animals, do not act by memorizing and replaying motions; they adapt as they go. Teaching a robot from human video is hard first for a mechanical reason, the mismatch between a human body and a robot's, and the gap between simulation and the real world [52]. But the mechanical gap is not the only one, and better data would not close it: a recorded action does not carry the judgment that produced it, the same gap as in the office, where the data holds the finished work and never the reasoning behind it. So a robot can learn from a million demonstrations and still miss what matters [53]: for work whose process is not standardized, unlike the welding long since fixed on the assembly line, what matters is judgment, the real-time kind that keeps an action within bounds in a world it has not seen. On a far smaller brain an ant or an octopus still out-manipulates the best humanoid, because the bottleneck is not just data but this adaptive judgment [12].

An independent 2026 audit finds the gap has not closed: robots still run three to ten times slower than people, their skills rarely transfer, and everyday jobs like cooking a meal or assembling flat-pack furniture remain at the early-research stage [54]. Nor does scaling the datacenter models improve the robot's control, because roboticists split that control to match the problem: a slow deliberative loop, System 2, for the plan and its judgment, and a fast reactive loop, System 1, for the reflex and its adaptation. The slow loop fails for the same reason as office work. The fast loop runs a small model at millisecond latency no datacenter can reach, out-adapted by an ant. Neither loop's work is cheaply checkable, so scaling helps neither. And in the physical world the check itself is expensive: a robot's error is not a wrong answer a test catches and discards, but an action already taken, its damage already real.

The skilled trades need both judgment and adaptation at once. A plumber diagnoses the fault, which is judgment, and fits the repair in an unstructured space every time, which is adaptation at its hardest. A person brings both, on call and costing little when idle, while a general machine must cover its cost whether or not there is work. That is why the trades rank as the least automatable work in nearly every forecast, and their numbers keep growing.

2.3 The cost of automation with a general-purpose robot

Automating physical work is expensive, and industrial robotics beat that cost only by constraining the world around the machine, with a fenced safe space, heavy calibration, and one repeated job. The car plant shows the consequences: the body shop, where panels are welded into a frame, has been the most automated stage since the 1980s and reduced employment where it was adopted most [1], while the final assembly on a floor next to it, fitting soft and varied trim as the line moves, is the least automated and still mostly manual [55].

The optimist's counterargument is the software one again: with enough work, a general-purpose robot could be engineered, trained, and guarded until it could do the work safely. But the guarding is itself judgment, the safety and reliability worked out in advance and built in before the machine can be trusted, the physical version of the harness: one design for every task and setting, impossible to build without a person in the loop. It is also what makes the whole process capital-intensive

and, on current evidence, unproven at open-world generality: even after years of development it succeeds only in a very narrow range of work. A humanoid in the open carries more of that guarding at lower utilization, not less, held by the collaborative-safety standards to a fraction of the speed and payload of a fenced industrial arm, so to match industrial work it would have to be caged after all [56]. Because the industrial robot could be fenced off, it never needed the human shape: built around the task instead, it came out cheaper than the worker it replaced. The humanoid gives that up, fixing the human form first.

2.4 The impact so far

The remaining jobs in the service industry and in physical labor are where automation has already arrived, and where it did, the workers were not displaced. Self-checkout is the clearest service case: it did not remove the labor but handed the scanning to the customer and kept staff for loss prevention, the produce, and the exceptions. Shrink wiped out the savings, and retailers are pulling back [57, 58].

Where the last wave of robots arrived to fill a shortage rather than cut a payroll, employment held or rose: developed economies are short of factory workers, with the United States alone projecting more than two million unfilled jobs by 2030 [59], and firms add robots to cover the gap [60]; in Japanese nursing homes robot adoption came alongside higher staff employment and retention [61].

Cost has already been cut to the minimum on both sides, the specialized machine and the flexible worker. An appliance is cheap because it is built by the million to do one job: a dishwasher costs a few hundred dollars. The cheapest humanoids now list around \$6,000 with none of that scale [62], and their price is falling fast, but so is the dishwasher's. The manual work those machines leave, loading the dishwasher, folding the laundry, emptying the bin, goes to a part-time cleaner paid only for the hours a household needs, about \$18 an hour [20]. On many factory lines the worker is already a modest share of the cost, with materials the larger part, so removing that worker barely lowers the total (Figure 3) [63].

A general robot has to beat both at once, the cheap specialized machine and the cheap, flexible worker, and earn back a heavy fixed cost whether it runs or sits idle. The human form was chosen to reach exactly this work, the open-world tasks no fixed machine could, but the shape was never what was missing: the work needs judgment and adaptation, and a human-shaped machine has neither. So it pays for its generality twice, costing more than the specialized tool at the work the tool already does, and still unable to supply the judgment and adaptation it was built to reach.

The winning robotic solution, then, is the specialized one: the built world was made for people using tools, so the machine that pays off is the one that becomes the tool, not the one that mimics the person. The battlefield, which no labor taxonomy covers and whose work spans planning, logistics, and fabrication at once, shows this at its clearest. The drone is a tool that flies itself, and Ukraine fielded around three million first-person-view drones in 2025, while the ground robots scaling up beside them were specialized crewed platforms, each fitted for one job, hauling supplies or evacuating the wounded, not a general machine that mimics a soldier [64–66]. In the factory the tool is the fixed arm; in the home it is the washer. The humanoid instead rebuilds the body and



Figure 3: On a factory line the worker is already a modest part of the cost.

then still has to pick the tool up.

3. What stays human

Decades ago, Polanyi held that we know more than we can tell, so the tacit part of skilled work resists codification [67]; Dreyfus argued that human competence rests on embodied know-how no formal rule can capture [68]; and Moravec named the mirror image, that the reasoning we find hard is easy to automate while the sensorimotor skill we find easy is not [11]. The task view in economics makes the same point, machines taking particular tasks rather than whole jobs, and finds the near-term effect real but bounded, only a fraction of tasks automatable at a profit [18, 25, 69, 70]. Each wave since, from expert systems to the large language models, automated more of what a cheap check can settle, and none has learned to judge well enough. The reason is still a hypothesis. Either way, judgment resists, and the harness to substitute for it stays expensive.

4. What is coming, and where the opportunities are

The expense of building the harness in software development bought a new capability: a higher-level language over code, as high-level languages were over assembly, with the model as the compiler [71]. Unlike a real compiler's target, this one has no precise semantics, which is why it needs the verification harness; and a tool that raises productivity like that does not disappear. What is left is engineering, making it cheaper and more reliable, and new projects and teams built around the tools capture the gains, rather than older ones retrofitted to them.

Fluency is cheap to produce and costly to trust, so the binding constraint is not generation but verification, and the models can be turned on the verification itself: in one industrial study an LLM screening static-analysis alarms cleared 94 to 98% of the false positives that were costing ten to twenty minutes of review each, at a fraction of the price [72]. The automatable frontier will also keep widening: imitation and reinforcement learning are lowering the cost of standing up specialized machines, and robotic process automation now augmented by AI absorbs the exceptions that used to break the old brittle scripts, so more of the routine becomes worth automating. This is the main way productivity rises outside software. But widening the frontier is not the same as displacing: lowering the cost of producing something, unless demand for it is inelastic, raises how much is consumed, so the effect is more output rather than fewer people, and only in saturated markets does cheaper production cut headcount.

The clearest openings are in the work a computer can check for itself. Formal verification, cryptographic proof, and the AI-assisted search that lately produced an explicit counterexample to the Jacobian conjecture [73, 74] all share the property that makes AI reliable: a person sets the exact objective, and any candidate answer can be checked. This is also why reasoning models trained

against verifiable rewards have advanced fastest in mathematics and code with tests: those are the places where a cheap checker exists, a test suite for code and a proof assistant for mathematics, which is evidence for the boundary, not against it. Tao makes the same point in mathematics: even granting that AI comes to solve research problems, the verified proof still leaves the understanding, exposition, and community acceptance to people, the stages he calls the slowest and least amenable to automation [75]. Several of these fields have been starved of skilled labor for decades; formal methods in particular have been held back not by lack of demand but by how few people can practice them [76], so most software, hardware, and protocols ship unverified. Lowering the cost of that checkable work does not displace a thin workforce; it clears a backlog never attempted for want of hands.

AlphaFold, one of AI's largest scientific wins, predicted the structures of more than 200 million proteins and retired years of laboratory grunt-work [77, 78], yet a predicted structure is a starting point, not a medicine; it still needs chemists to confirm it, trials to prove it, and factories to make it [79]. The breakthrough moved the work downstream; it did not end it.

5. A near-term forecast

Forecasts of quick, wholesale displacement of labor through automation have a mixed record, which is a caution for this one too. Frey and Osborne's 47% overshoot, as did IBM's Watson in oncology and the sewing robots promised to empty the garment factories [19, 80, 81], while the same expert class missed deep learning and large language models almost entirely. Blinder's 2007 estimate is the sharper case: he judged 22 to 29% of US jobs potentially offshorable, later independently corroborated [82, 83], yet the wholesale displacement that potential implied never came at that scale, even as offshoring did reshape particular sectors, because a potential is not a realization. Even the forecast's loudest voices have qualified it: in 2026 Sam Altman said he had expected more entry-level displacement by now than has happened, and Dario Amodei allowed that Jevons dynamics, more tasks per job, could offset the job cuts he had warned of [84, 85]. This paper is itself a forecast, and subject to the same caution: its bet is not that capability stalls, the error the failed forecasts made, but that economic diffusion lags capability over three to five years.

The AI inference buildout is a vast bet, part of it on displacement. The forecasts that price it are large, some 300 million jobs exposed globally [86] or 92 million displaced by 2030 [87], and the compute commitments signed against them already run to a trillion dollars or more, booked as multi-year backlog that dwarfs current lab revenues [88]. A cost compounding at triple digits cannot be met from the IT budget, a few percent of revenue; only payroll, several times larger, is a pool deep enough to justify it. So the demand case rests in part on AI displacing enough workers to free the wages that would pay for the compute, and in part on augmentation proving valuable enough that firms sharply raise what they spend on AI. But the productivity that does materialize, output per hour up 2.2% in 2025 as hours stayed roughly flat [89], accrues to the firms using AI, not the labs paying for it. The humanoid wave, sometimes offered as the alternative buyer, pays off around 2035, past this horizon and drawing on the same capital. The full arithmetic is beyond this paper's scope; the point is only that a large part of the buildout is priced on a displacement that, on the mechanism above, may not occur in time. Financial pressure to displace is not the capability to

displace: a firm that cuts judgment-heavy labor to service the buildout sheds work it still needs, the failure this paper describes, not a way around it.

For a leader, the horizon that matters is three to five years, and on it there is no present basis for large displacement. Over a longer horizon this could change, if models learn to carry skills across tasks or labor grows scarce, and the claim is falsifiable: it fails if, within this horizon, employment declines materially in occupations dominated by non-standardized judgment work, the senior and mid-career roles, not only in the entry-level, standardized layer. Short of that, cutting labor on AI's promise sheds roles the business still needs and stakes more than the AI budget on a false premise. Near-term, the tools augment the same headcount rather than replacing it, so the firm pays for both the wages and the AI. Worse, it breaks the apprenticeship, hollowing the pipeline where the next senior trains alongside someone who already has the judgment the machine cannot supply.

References

- [1] Daron Acemoglu and Pascual Restrepo. Robots and Jobs: Evidence from US Labor Markets. *Journal of Political Economy*, 128(6):2188–2244, June 2020. ISSN 0022-3808, 1537-534X. doi: 10.1086/705716. URL <https://www.journals.uchicago.edu/doi/10.1086/705716>.
- [2] Alan S. Blinder. *Offshoring: big deal, or business as usual*. Center for Economic Policy Studies, Princeton University, 2007. URL <https://www.princeton.edu/~ceps/workingpapers/149blinder.pdf>.
- [3] Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, and Alex Ray. Training language models to follow instructions with human feedback. *Advances in neural information processing systems*, 35: 27730–27744, 2022. URL <https://proceedings.neurips.cc/paper/2022/hash/b1efde53be364a73914f58805a001731-Abstract.html>.
- [4] Linda Flower and John R. Hayes. A Cognitive Process Theory of Writing. *College Composition & Communication*, 32(4):365–387, December 1981. ISSN 0010-096X, 1939-9006. doi: 10.58680/cc198115885. URL <https://publicationsncte.org/content/journals/10.58680/cc198115885>.
- [5] Carl Bereiter and Marlene Scardamalia. *The psychology of written composition*. Routledge, 2013. URL <https://api.taylorfrancis.com/content/books/mono/download?identifierName=doi&identifierValue=10.4324/9780203812310&type=googlepdf>.
- [6] Chunting Zhou, Pengfei Liu, Puxin Xu, Srinivasan Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, and Lili Yu. Lima: Less is more for alignment. *Advances in Neural Information Processing Systems*, 36:55006–55021, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/ac662d74829e4407ce1d126477f4a03a-Abstract-Conference.html.
- [7] Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, D. D. L. Casas, Lisa Anne Hendricks, Johannes Welbl, and Aidan Clark. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 10, 2022. URL <https://proceedings.neurips.cc/paper/2022/file/c1e2faff6f588870935f114ebe04a3e5-Paper-Conference.pdf>.

- [8] Epoch AI. Data on AI Models. URL <https://epoch.ai/data/ai-models>.
- [9] Mrinank Sharma, Meg Tong, Tomek Korbak, David Duvenaud, Amanda Asbell, Sam Bowman, Esin Durmus, Zac Hatfield-Dodds, Scott Johnston, and Shauna Kravec. Towards understanding sycophancy in language models. In *International Conference on Learning Representations*, volume 2024, pages 110–144, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/0105f7972202c1d4fb817da9f21a9663-Abstract-Conference.html.
- [10] Ethan Perez, Sam Ringer, Kamile Lukosiute, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, and Saurav Kadavath. Discovering language model behaviors with model-written evaluations. In *Findings of the association for computational linguistics: ACL 2023*, pages 13387–13434, 2023. URL <https://aclanthology.org/2023.findings-acl.847/>.
- [11] Hans Moravec. *Mind children: The future of robot and human intelligence*. Harvard University Press, 1988. URL <https://books.google.com/books?hl=en&lr=&id=56mb7XuSx3QC&oi=fnd&pg=PA1&dq=Mind+Children:+The+Future+of+Robot+and+Human+Intelligence.+&ots=iltHDVJHxy&sig=UovnhWSX4h50WMzXnLqbHYV2hNU>.
- [12] Jinda Cui and Jeff Trinkle. Toward next-generation learned robot manipulation. *Science Robotics*, 6(54):eabd9461, May 2021. ISSN 2470-9476. doi: 10.1126/scirobotics.abd9461. URL <https://www.science.org/doi/10.1126/scirobotics.abd9461>.
- [13] Edsger W. Dijkstra. On the foolishness of “natural language programming”. In Friedrich L. Bauer, Manfred Broy, E. W. Dijkstra, S. L. Gerhart, D. Gries, M. Griffiths, J. V. Guttag, J. J. Horning, S. S. Owicki, C. Pair, H. Partsch, P. Pepper, M. Wirsing, and H. Wössner, editors, *Program Construction*, volume 69, pages 51–53. Springer Berlin Heidelberg, Berlin, Heidelberg, 1979. ISBN 978-3-540-09251-3 978-3-540-35312-6. doi: 10.1007/BFb0014656. URL <https://link.springer.com/10.1007/BFb0014656>. Series Title: Lecture Notes in Computer Science.
- [14] Ajay Agrawal, Joshua Gans, and Avi Goldfarb. *Prediction machines, updated and expanded: The simple economics of artificial intelligence*. Harvard Business Press, 2022. URL <https://books.google.com/books?hl=en&lr=&id=8MBYEAAAQBAJ&oi=fnd&pg=PT6&dq=prediction+machines&ots=0ELbboODgr&sig=w4tuSsLCtD-YvorUQpCq7G1MAXk>.
- [15] R. M. Solow. “We’d Better Watch Out”, New York Times Book Review, July 12. 1987.
- [16] Stephen D. Oliner and Daniel E. Sichel. The resurgence of growth in the late 1990s: is information technology the story? *Journal of economic perspectives*, 14(4):3–22, 2000. URL https://www.aeaweb.org/articles?id=10.1257%2Fjep.14.4.3&utm_source=chatgpt.com.
- [17] Erik Brynjolfsson and Lorin M. Hitt. Beyond computation: Information technology, organizational transformation and business performance. *Journal of Economic perspectives*, 14(4):23–48, 2000. URL <https://www.aeaweb.org/articles?id=10.1257/jep.14.4.23>.
- [18] David H. Autor, Frank Levy, and Richard J. Murnane. The skill content of recent technological change: An empirical exploration. *The Quarterly journal of economics*, 118(4):1279–1333, 2003. URL <https://academic.oup.com/qje/article-abstract/118/4/1279/1925105>.

- [19] Carl Benedikt Frey and Michael A. Osborne. The future of employment: How susceptible are jobs to computerisation? *Technological forecasting and social change*, 114:254–280, 2017. URL <https://www.sciencedirect.com/science/article/pii/S0040162516302244>.
- [20] Bureau of Labor Statistics. Occupational Employment and Wage Statistics (OEWS) Tables, 2025. URL <https://www.bls.gov/oes/tables.htm>.
- [21] Duncan G. Copeland and James L. McKenney. Airline reservations systems: lessons from history. *MIS quarterly*, 12(3):353–370, 1988. URL <https://misq.umn.edu/misq/article/12/3/353/988>.
- [22] Susan Wharton Gates, Vanessa Gail Perry, and Peter M. Zorn. Automated underwriting in mortgage lending: Good news for the underserved? *Housing Policy Debate*, 13(2):369–391, January 2002. ISSN 1051-1482, 2152-050X. doi: 10.1080/10511482.2002.9521447. URL <http://www.tandfonline.com/doi/abs/10.1080/10511482.2002.9521447>.
- [23] Maura R. Grossman and Gordon V. Cormack. Technology-assisted review in e-discovery can be more effective and more efficient than exhaustive manual review. *Rich. JL & Tech.*, 17:1, 2010. URL <https://heinonline.org/hol-cgi-bin/get.pdf.cgi?handle=hein.journals/jolt17§ion=22>.
- [24] Terrence Hendershott, Charles M. Jones, and Albert J. Menkveld. Does Algorithmic Trading Improve Liquidity? *The Journal of Finance*, 66(1):1–33, February 2011. ISSN 0022-1082, 1540-6261. doi: 10.1111/j.1540-6261.2010.01624.x. URL <https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6261.2010.01624.x>.
- [25] Matthias Mertens, Adam Kuzee, Brittany S. Harris, Harry Lyu, Wensu Li, Jonathan Rosenfeld, Meiri Anto, Martin Fleming, and Neil Thompson. Crashing Waves vs. Rising Tides: Findings on AI Automation from Thousands of Worker Evaluations of Labor Market Tasks, July 2026. URL <http://arxiv.org/abs/2604.01363>. arXiv:2604.01363 [cs.AI].
- [26] National Center for O*NET Development. O*NET OnLine. URL <https://www.onetonline.org/>.
- [27] Fabrizio Dell’Acqua, Edward McFowland III, Ethan R. Mollick, Hila Lifshitz-Assaf, Katherine Kellogg, Saran Rajendran, Lisa Kraye, François Candelon, and Karim R. Lakhani. Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality. *Harvard business school technology & operations mgt. Unit working paper*, (24-013), 2023. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4573321.
- [28] Erik Brynjolfsson. The Turing Trap: The Promise & Peril of Human-Like Artificial Intelligence. *Journal of the American Academy of Arts & Sciences*, 272, 2022. URL https://www.jstor.org/stable/pdf/48662041.pdf?refreqid=excelsior%3A1ff6fd4b0a72f3de6413b3c9de810505&ab_segments=&origin=&initiator=&acceptTC=1.

- [29] Kunal Handa, Alex Tamkin, Miles McCain, Saffron Huang, Esin Durmus, Sarah Heck, Jared Mueller, Jerry Hong, Stuart Ritchie, Tim Belonax, Kevin K. Troy, Dario Amodei, Jared Kaplan, Jack Clark, and Deep Ganguli. Which Economic Tasks are Performed with AI? Evidence from Millions of Claude Conversations, February 2025. URL <http://arxiv.org/abs/2503.04761>. arXiv:2503.04761 [cs.CY].
- [30] Erik Brynjolfsson, Danielle Li, and Lindsey Raymond. Generative AI at work. *The Quarterly Journal of Economics*, 140(2):889–942, 2025. URL <https://academic.oup.com/qje/article-abstract/140/2/889/7990658>.
- [31] Shakked Noy and Whitney Zhang. Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, 381(6654):187–192, July 2023. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.adh2586. URL <https://www.science.org/doi/10.1126/science.adh2586>.
- [32] Sida Peng, Eirini Kalliamvakou, Peter Cihon, and Mert Demirer. The Impact of AI on Developer Productivity: Evidence from GitHub Copilot, February 2023. URL <http://arxiv.org/abs/2302.06590>. arXiv:2302.06590 [cs.SE].
- [33] Joel Becker, Nate Rush, Elizabeth Barnes, and David Rein. Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity, July 2025. URL <http://arxiv.org/abs/2507.09089>. arXiv:2507.09089 [cs.AI].
- [34] Yue Liu, Ratnadira Widayarsi, Yanjie Zhao, Ivana Clairine Irsan, Junkai Chen, and David Lo. Debt Behind the AI Boom: A Large-Scale Empirical Study of AI-Generated Code in the Wild, April 2026. URL <http://arxiv.org/abs/2603.28592>. arXiv:2603.28592 [cs.SE].
- [35] Lisanne Bainbridge. Ironies of automation. *Automatica*, 19(6):775–779, November 1983. ISSN 0005-1098. doi: 10.1016/0005-1098(83)90046-8. URL <https://www.sciencedirect.com/science/article/pii/0005109883900468>.
- [36] Raja Parasuraman and Dietrich H. Manzey. Complacency and Bias in Human Use of Automation: An Attentional Integration. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 52(3):381–410, June 2010. ISSN 0018-7208, 1547-8181. doi: 10.1177/0018720810376055. URL <https://journals.sagepub.com/doi/10.1177/0018720810376055>.
- [37] Andrew M. Bean, Ryan Othniel Kearns, Angelika Romanou, Franziska Sofia Hafner, Harry Mayne, Jan Bartzner, Negar Foroutan Eghlidi, Chris Schmitz, Karolina Korgul, and Hunar Batra. Measuring what matters: Construct validity in large language model benchmarks. *Advances in Neural Information Processing Systems*, 38, 2026. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/1967e0fc3aa6cbbace562f5cb8e3954e-Abstract-Datasets_and_Benchmarks_Track.html.
- [38] Anthropic. How Anthropic enables self-service data analytics with Claude | Claude by Anthropic. URL <https://claude.com/blog/how-anthropic-enables-self-service-data-analytics-with-claude>.

- [39] Longju Bai, Zhemin Huang, Xingyao Wang, Jiao Sun, Rada Mihalcea, Erik Brynjolfsson, Alex Pentland, and Jiaxin Pei. How Do AI Agents Spend Your Money? Analyzing and Predicting Token Consumption in Agentic Coding Tasks, April 2026. URL <http://arxiv.org/abs/2604.22750>. arXiv:2604.22750 [cs.CL].
- [40] Artificial Analysis Intelligence Index. URL <https://artificialanalysis.ai/evaluations/artificial-analysis-intelligence-index>.
- [41] Ivan Yotzov, Jose Maria Barrero, Nicholas Bloom, Philip Bunn, Steven J. Davis, Kevin M. Foster, Aaron Jalca, Brent H. Meyer, Paul Mizen, Michael A. Navarrete, Pawel Smietanka, Gregory Thwaites, and Ben Zhe Wang. Firm Data on AI, February 2026. URL <https://www.nber.org/papers/w34836>.
- [42] Anders Humlum and Emilie Vestergaard. *Large language models, small labor market effects*, volume 33777. National Bureau of Economic Research Cambridge, MA, 2025. URL https://www.nber.org/system/files/working-papers/w33777/w33777.pdf?mc_cid=2bd0264926&mc_eid=976bc2a5b6.
- [43] Aditya Challapally, Chris Pease, Ramesh Raskar, and Pradyumna Chari. STATE OF AI IN BUSINESS 2025. 2025. URL https://mlq.ai/media/quarterly_decks/v0.1_State_of_AI_in_Business_2025_Report.pdf.
- [44] Melanie Arntz, Terry Gregory, and Ulrich Zierahn. The risk of automation for jobs in OECD countries: A comparative analysis. *OECD social, employment and migration working papers*, 2016, 2016. URL https://www.oecd.org/en/publications/the-risk-of-automation-for-jobs-in-oecd-countries_5jlz9h56dvq7-en.html?utm_source=chatgpt.com.
- [45] Erik Brynjolfsson, Bharat Chandar, and Ruyu Chen. Canaries in the Coal Mine?: Six Facts about the Recent Employment Effects of Artificial Intelligence. 2025. URL https://digitaleconomy.stanford.edu/app/uploads/2025/12/CanariesintheCoalMine_Nov25.pdf.
- [46] Maxim Massenkoff and Peter McCrory. Labor market impacts of AI: A new measure and early evidence, 2026. URL <https://www.anthropic.com/research/labor-market-impacts>.
- [47] William Rodgers and Alice Kassens. How Shifts in Labor Supply and Demand Shape Outcomes for Young Workers, 2026. URL <https://www.stlouisfed.org/on-the-economy/2026/jun/how-shifts-labor-supply-demand-shape-outcomes-young-workers>.
- [48] Morgan R. Frank, Alireza Javadian Sabet, Lisa Simon, Sarah H. Bana, and Renzhe Yu. AI-exposed jobs deteriorated before ChatGPT, January 2026. URL <http://arxiv.org/abs/2601.02554>. arXiv:2601.02554 [econ.GN].
- [49] Jeremy Greenwood, Ananth Seshadri, and Mehmet Yorukoglu. Engines of liberation. *The Review of economic studies*, 72(1):109–133, 2005. URL <https://academic.oup.com/restud/article-abstract/72/1/109/1581345>.

- [50] International Federation of Robotics. Record of 4 Million Robots in Factories Worldwide, 2024. URL <https://ifr.org/ifr-press-releases/news/record-of-4-million-robots-working-in-factories-worldwide>.
- [51] International Federation of Robotics. US Robot Industry Returns to Double Digit Growth, 2026. URL <https://ifr.org/ifr-press-releases/news/us-robot-industry-returns-to-double-digit-growth>.
- [52] Tyler Ga Wei Lum, Olivia Y. Lee, C. Karen Liu, and Jeannette Bohg. Crossing the Human-Robot Embodiment Gap with Sim-to-Real RL using One Human Demonstration, August 2025. URL <http://arxiv.org/abs/2504.12609>. arXiv:2504.12609 [cs.RO].
- [53] Abby O'Neill, Abdul Rehman, Abhiram Maddukuri, Abhishek Gupta, Abhishek Padalkar, Abraham Lee, Acorn Pooley, Agrim Gupta, Ajay Mandlekar, and Ajinkya Jain. Open x-embodiment: Robotic learning datasets and rt-x models: Open x-embodiment collaboration. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 6892–6903. IEEE, 2024. URL <https://ieeexplore.ieee.org/abstract/document/10611477/>.
- [54] Yann Rivière and Jean-Stanislas Denain. Where Autonomy Works: Evaluating Robot Capabilities in 2026. URL <https://epoch.ai/publications/where-autonomy-works-evaluating-robot-capabilities-in-2026>.
- [55] Austin Weber. Beyond the Welding Line: Robots and Automotive Assembly, 2022. URL <https://www.assemblymag.com/articles/96973-beyond-the-welding-line-robots-and-automotive-assembly>.
- [56] International Organization for Standardization. ISO/TS 15066:2016, 2016. URL <https://www.iso.org/standard/62996.html>.
- [57] Nathaniel Meyersohn. Dollar General will remove self-checkout from 300 stores to prevent shoplifting, March 2024. URL <https://www.cnn.com/2024/03/14/business/self-checkout-dollar-general>.
- [58] Associated Press. Amazon says it is removing Just Walk Out technology from its Fresh grocery stores. *The Associated Press*, April 2024. URL <https://www.npr.org/2024/04/03/1242448552/amazon-remove-walk-out-technology>.
- [59] Heather Ashton, Chad Moutray, and Victor Reyes. Creating pathways for tomorrow's workforce today, 2021. URL <https://www.deloitte.com/us/en/insights/industry/manufacturing-industrial-products/manufacturing-industry-diversity.html>.
- [60] Daron Acemoglu and Pascual Restrepo. Demographics and automation. *The Review of Economic Studies*, 89(1):1–44, 2022. URL <https://academic.oup.com/restud/article-abstract/89/1/1/6295889>.
- [61] Karen Eggleston, Yong Suk Lee, and Toshiaki Iizuka. Robots and labor in the service sector: Evidence from nursing homes. Technical report, National Bureau of Economic Research, 2021. URL <https://www.nber.org/papers/w28322>.

- [62] Eugene Demaitre. Unitree designs R1 humanoid robot to be agile and affordable, July 2025. URL <https://www.therobotreport.com/unitree-designs-r1-humanoid-robot-to-be-agile-and-affordable/>.
- [63] U.S. Census Bureau. Annual Survey of Manufactures (ASM). URL <https://www.census.gov/programs-surveys/asm.html>.
- [64] Tim Zadorozhnyy. Ukraine on track to receive total of 3 million FPV drones in 2025, defense minister says, December 2025. URL <https://kyivindependent.com/ukraine-on-track-to-receive-total-of-3-million-fpv-drones-in-2025/>.
- [65] Jorge Rivero. Networked for War: Lessons from Ukraine’s Ground Robots, March 2026. URL <https://mwi.westpoint.edu/networked-for-war-lessons-from-ukraines-ground-robots/>.
- [66] Tania Myronyshena. Ukraine’s new ground drones are hitting the battlefield in ever-increasing numbers, July 2025. URL <https://kyivindependent.com/ukraine-approves-new-ground-robots-to-relieve-infantry-strained-by-mobilization-crisis/>.
- [67] Michael Polanyi. The tacit dimension. In *Knowledge in organisations*, pages 135–146. Routledge, 2009. URL <https://api.taylorfrancis.com/content/chapters/edit/download?identifierName=doi&identifierValue=10.4324/9780080509822-7&type=chapterpdf>.
- [68] Hubert L. Dreyfus. *What Computers Still Can’t Do: A Critique of Artificial Reason*. The MIT Press, Cambridge, Mass., 2009. ISBN 978-0-262-54067-4.
- [69] David H. Autor. Why are there still so many jobs? The history and future of workplace automation. *Journal of economic perspectives*, 29(3):3–30, 2015. URL <https://www.aeaweb.org/articles?id=10.1257%2Fjep.29.3.3>.
- [70] Daron Acemoglu. The Simple Macroeconomics of AI. (WP 32487), 2024. URL <http://www.nber.org/papers/w32487>.
- [71] Shuyuan Xu, Zelong Li, Kai Mei, and Yongfeng Zhang. AIOS Compiler: LLM as Interpreter for Natural Language Programming and Flow Programming of AI Agents, May 2024. URL <http://arxiv.org/abs/2405.06907>. arXiv:2405.06907 [cs.CL].
- [72] Xueying Du, Jiayi Feng, Yi Zou, Wei Xu, Jie Ma, Wei Zhang, Sisi Liu, Xin Peng, and Yiling Lou. Reducing False Positives in Static Bug Detection with LLMs: An Empirical Study in Industry, January 2026. URL <http://arxiv.org/abs/2601.18844>. arXiv:2601.18844 [cs.SE].
- [73] Terence Tao. A digestion of the Jacobian conjecture counterexample, July 2026. URL <https://terrytao.wordpress.com/2026/07/21/a-digestion-of-the-jacobian-conjecture-counterexample/>.
- [74] Aaron Lou. Jacobian Conjecture Prompt. 2026. URL https://aaronlou.com/jacobian_counterexample_prompt.pdf.
- [75] Terence Tao. Mathematics in the age of AI - Public lecture, International Congress of Mathematicians 2026. 2026. URL <https://teorth.github.io/tao-web/slides/age-of-ai-icm-2026.pdf>.

- [76] Maurice H. ter Beek, Rod Chapman, Rance Cleaveland, Hubert Garavel, Rong Gu, Ivo ter Horst, Jeroen J. A. Keiren, Thierry Lecomte, Michael Leuschel, Kristin Yvonne Rozier, Augusto Sampaio, Cristina Seceleanu, Martyn Thomas, Tim A. C. Willemse, and Lijun Zhang. Formal Methods in Industry. *Formal Aspects of Computing*, 37(1):7:1–7:38, December 2024. ISSN 0934-5043. doi: 10.1145/3689374. URL <https://dl.acm.org/doi/10.1145/3689374>.
- [77] John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, and Anna Potapenko. Highly accurate protein structure prediction with AlphaFold. *nature*, 596(7873):583–589, 2021. URL <https://www.nature.com/articles/s41586-021-03819-2>.
- [78] Mihaly Varadi, Damian Bertoni, Paulyna Magana, Urmila Paramval, Ivanna Pidruchna, Malarvizhi Radhakrishnan, Maxim Tsenkov, Sreenath Nair, Milot Mirdita, and Jingi Yeo. AlphaFold Protein Structure Database in 2024: providing structure coverage for over 214 million protein sequences. *Nucleic acids research*, 52(D1):D368–D375, 2024. URL <https://academic.oup.com/nar/article-abstract/52/D1/D368/7337620>.
- [79] David Autor, Caroline Chin, Anna Salomons, and Bryan Seegmiller. New frontiers: The origins and content of new work, 1940–2018. *The Quarterly Journal of Economics*, 139(3):1399–1465, 2024. URL <https://academic.oup.com/qje/article-abstract/139/3/1399/7630187>.
- [80] Casey Ross and Ike Swetlitz. IBM’s Watson supercomputer recommended ‘unsafe and incorrect’ cancer treatments, internal documents show. *Stat*, 25:1–10, 2018. URL <https://www.statnews.com/wp-content/uploads/2018/09/IBMs-Watson-recommended-unsafe-and-incorrect-cancer-treatments-STAT.pdf>.
- [81] Fernanda Bárcia de Mattos, Jeff Eisenbraun, David Kucera, and Arianna Rossi. Automation, employment and reshoring in the apparel industry: Long-term disruption or a storm in a teacup?, May 2020. URL <https://www.ilo.org/publications/automation-employment-and-reshoring-apparel-industry-long-term-disruption>.
- [82] Alan S Blinder. How Many U.S. Jobs Might Be Offshorable? (CEPS Working Paper 142), 2007.
- [83] Troy D. Smith and Jan W. Rivkin. A Replication Study of Alan Blinder’s ‘How Many US Jobs Might Be Offshorable?’. *Harvard Business School Strategy Unit Working Paper*, (08-104), 2008. URL https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1148273.
- [84] Sasha Rogelberg. Sam Altman and Dario Amodei are both walking back their AI jobs apocalypse prophecies as they eye blockbuster IPOs, 2026. URL <https://fortune.com/2026/05/26/sam-altman-dario-amodei-walking-back-ai-jobs-apocalypse-prophecies-ipo/>.
- [85] Nick Lichtenberg. Dario Amodei spent last year warning of an AI white-collar bloodbath. Now he’s changing the narrative, 2026. URL <https://fortune.com/2026/05/05/dario-amodei-jevons-paradox-will-ai-wipe-out-white-collar-jobs/>.
- [86] Jan Hatzius, Joseph Briggs, Devesh Kodnani, and Giovanni Pierdomenico. The Potentially Large Effects of Artificial Intelligence on Economic Growth, 2023. URL <https://www.gspublis>

hing.com/content/research/en/reports/2023/03/27/d64e052b-0f6e-45d7-967b-d7be35fabd16.html.

- [87] World Economic Forum. The Future of Jobs Report 2025, 2025. URL <https://www.weforum.org/publications/the-future-of-jobs-report-2025/>.
- [88] George Lee and Lucas Greenbaum. Tracking Trillions: The Assumptions Shaping the Scale of the AI Build-Out, 2026. URL <https://www.goldmansachs.com/insights/articles/tracking-trillions-the-assumptions-shaping-scale-of-the-ai-build-out>.
- [89] U.S. Bureau of Labor Statistics. PRODUCTIVITY AND COSTS Fourth Quarter and Annual Averages 2025, Preliminary, 2026. URL <https://www.bls.gov/productivity/>.